

1. Administrative & Study Identification

Full Study Title: A Phase I/II Open-Label Study to Evaluate the Safety, Cellular Kinetics and Efficacy of AZD5851, a Chimeric Antigen Receptor T-Cell CAR-T Therapy Directed Against GPC3 in Adult Participants With Advanced/Recurrent HEpatocellular CarciNomA: ATHENA **Short Title/Acronym:** ATHENA **Protocol Number:** D7670C00001 **NCT Number:** NCT06084884 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD5851 (Chimeric Antigen Receptor T-Cell / CAR-T Therapy directed against GPC3) **Phase of Development:** Phase I/II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective: To evaluate the safety, tolerability, and antitumour activity (efficacy) of AZD5851 in patients with GPC3+ advanced/recurrent hepatocellular carcinoma (HCC). **Secondary Objectives:** To evaluate objective response rate (ORR), best overall response (BOR), duration of response (DoR), disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and cellular kinetics/pharmacokinetics.

2.2 Background & Rationale To address the significant unmet therapeutic need for patients with advanced/recurrent HCC who progress after first-line treatments (such as checkpoint inhibitor therapy). Glypican-3 (GPC3) is highly overexpressed by tumor cells in HCC and virtually absent in healthy tissues, making it an ideal target for CAR-T therapy. AZD5851 is an autologous CAR-T product that expresses a CAR specific for GPC3, armed with a dominant negative TGFβRII to protect against immunosuppressive tumor microenvironments. This study aims to evaluate the viability of this adoptive cellular immunotherapy as a novel treatment option.

3. Study Design & Methodology

3.1 Design Framework

Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Dose escalation and expansion)

3.2 Planned Enrollment & Duration

Target \$N\$: Approximately 94 evaluable patients required **Number of Sites:** Multicenter (Global/US sites including Mayo Clinic, City of Hope, UCSF, KU Cancer Center, etc.) **Estimated Study Start:** 14-Dec-2023 **Estimated Primary Completion:** 13-Dec-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years, having voluntarily agreed to participate by giving written informed consent. 2. Confirmed advanced/recurrent or metastatic and/or unresectable HCC based on histopathological findings, with completed or intolerable prior standard systemic therapy. 3. GPC3-positive tumour as determined by a central laboratory using an analytically validated IHC assay; Barcelona Clinic Liver Cancer Stage B or C prior to apheresis, and Child-Pugh score Grade A.

4.2 Exclusion Criteria 1. Active or prior documented gastrointestinal (GI) variceal bleed or history of upper GI bleeding, ulcers, or esophageal varices with bleeding within 12 months. 2. Prior treatment with any CAR-T therapy directed at any target or any therapy that is targeted to GPC3. 3. History of liver transplantation or on a waiting list; or current clinically significant ascites.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency: Single intravenous (IV) infusion of AZD5851 following 3 consecutive doses of lymphodepleting chemotherapy (fludarabine and cyclophosphamide). **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Single administration followed by intensive short-term monitoring and long-term follow-up.

5.2 Concomitant & Prohibited Therapy

Permitted: Bridging therapy is permitted for disease control during the production of AZD5851 peripheral blood mononuclear cells (PBMCs). **Prohibited:** Receipt of the last dose of anticancer therapy within 5 half-lives or ≤ 21 days (whichever is shortest); chronic or recurrent severe autoimmune/immune-mediated disease requiring steroids or immune-suppressive treatments.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Pre-apheresis | Informed Consent, Medical History, Central Lab GPC3 IHC Testing, Leukapheresis (PBMC isolation) | | **Baseline** | Day -X to 0 | Lymphodepleting

Chemotherapy, Single dose AZD5851 IV Infusion | | **Interim** | Post-Infusion | Safety Labs, Cellular Kinetics, Efficacy Assessment (RECIST 1.1) | | **End of Study** | Disease Progression/Completion | Final Vital Signs, Exit Interview, Long-term Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Adverse Events (AEs), Dose-Limiting Toxicities (DLTs), and determination of the recommended Phase 2 dose to evaluate safety and tolerability.

Measurement Method: Clinical assessments and laboratory evaluations graded per standard toxicity criteria.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR), Duration of Response (DoR), and Progression-Free Survival (PFS) per RECIST 1.1 criteria. * Overall Survival (OS) and Cellular Kinetics/Pharmacokinetics of AZD5851.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring, with specific focus on CAR-T related toxicities like Cytokine Release Syndrome (CRS) and neurotoxicity. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External safety review committee assessing dose-escalation cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy, Safety Set for adverse events and cellular kinetics. **Sample Size**

Justification: $N = 94$; calculated to adequately evaluate safety, determine the maximum tolerated dose (MTD) during the dose-escalation phase, and detect preliminary antitumour activity in the dose-expansion phase. **Handling of Missing Data:** Standard oncological Phase I/II methodology (e.g., censoring at last known alive date for time-to-event endpoints like PFS/OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Intensive onsite monitoring during the peri-infusion period, followed by scheduled remote and onsite follow-ups. **Audit Trail:** Robust tracking of eCRF, centralized GPC3 IHC assay records, and chain-of-identity/chain-of-custody for autologous cell product manufacturing.

11. Study Identifiers & Governance

Primary ID: {{D7670C00001}} **Registry Number:** {{NCT06084884}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{ATHENA Trial}} **Official Regulatory Title:** {{A Phase I/II Open-Label Study to Evaluate the Safety, Cellular Kinetics and Efficacy of AZD5851, a Chimeric Antigen Receptor T-Cell CAR-T Therapy Directed Against GPC3 in Adult Participants With Advanced/Recurrent HEPatocellular CarciNomA: ATHENA}}

12. Clinical Framework (Hepatocellular Carcinoma Specific)

Primary Condition: {{Advanced/Recurrent Hepatocellular Carcinoma (HCC)}} **Diagnosis Threshold:** {{GPC3-positive tumour and Child-Pugh score Grade A}} **Medical Methodology:** {{Adoptive Cellular Immunotherapy (CAR-T)}} **Optimization Target:** {{T-cell mediated targeting and destruction of GPC3+ tumor cells}}

13. Study Procedures & Methodology

Management Pathway: {{Leukapheresis, bridging therapy, lymphodepletion, and CAR-T infusion}} **Education Protocol (HCP):** {{CAR-T administration, CRS, and neurotoxicity management training}} **Education Protocol (Patient):** {{Informed consent regarding autologous cell therapy risks and long-term follow up}} **Quality Audit Mechanism:** {{Central laboratory IHC testing for GPC3 status confirmation}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to study completion / Primary completion expected Dec 2027}} **Onsite Visit Cadence:** {{Intensive initial post-infusion monitoring, then transitioning to standard interval oncology follow-ups}} **Remote Monitoring Frequency:** {{As needed per protocol for long-term survival status}} **Checklist Review Interval:** {{Coinciding with RECIST 1.1 tumor evaluation scans}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				